

THE KOSELLECK MACHINE

Pipeline Manual

What each of the eight pipeline stages does internally,
why it is built the way it is, and where it still falls short of
what it should do.

REPOSITORY `koselleck-networks`

DOCUMENT `docs/pipeline_manual.tex` → `docs/pipeline_manual.pdf`

COMPANION `docs/overview.pdf` (setup and run commands), `docs/method.pdf` (the scientific argument)

BRANCH `corpus-rebuild`

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1 What this document is

`docs/overview.pdf` tells you what this project is and how to install and run it. `docs/method.pdf` argues the scientific case, with the resolution sweep and the labeling prompt reproduced in full. `README.md` in the repository root carries the same run commands as `docs/overview.pdf` for a reader who only has GitHub open. None of those three walks through what each stage of the pipeline does inside, or why it is built that way and not some simpler alternative. This document is that walkthrough, stage by stage, in run order: parse, repair OCR, bucket into periods, train embeddings, build the network, detect communities, label communities, measure change.

Two things apply across several stages, worth stating once instead of eight times. Every stage is a standalone script under `src/` that reads its input from the previous stage's output directory. Two of them, training embeddings and building the network, skip a variant whose output file already exists instead of overwriting it **a script has to be told to delete the file before it will redo the work: retraining a period's embeddings or rebuilding its network is expensive, and an accidental rerun should not throw that away for free**. The other five stages always overwrite. Second, most stages loop not only over the twenty period slices but over every **variant**: the combined corpus, plus one independently trained pass per region (American, British) found on disk. §10 explains what a region contains and why the project keeps them apart.

Where a stage still has an open problem, this document says so.

2 Stage 1: Parse

`src/parse_tcp.py` turns raw source files into one flat `processed/all_docs.jsonl`, one JSON record per document: `region`, `source`, `doc_id`, `year`, `text`. Bucketing into period windows is a separate, later stage (`bucket_periods.py`), so a period-boundary change ordinarily only means re-sorting text this stage already extracted, not rerunning it. OCR repair works differently: it runs inside this script, on text read directly from the raw archives, so a changed OCR rule does mean rerunning `parse_tcp.py` (§3).

Two source formats feed the same output file. The first is the Text Creation Partnership's P4 XML: TCP marks up EEBO, ECCO, and Evans by hand, and `extract_text` walks the XML tree collecting text between block-level tags (paragraphs, divisions, line breaks) while treating inline editorial tags, an illegible letter marked `<GAP/>` for instance, as invisible, not as word boundaries. Getting this distinction wrong once produced whole communities of nothing but split-word fragments, which is why the tag list in `BLOCK_TAGS` is spelled out rather than inferred. The publication date comes from the bibliographic `PUBLICATIONSTMT` under `SOURCEDESC`, not the sibling field under `FILEDESC`, which records

when TCP processed the file, not when the book was printed. A document with no extractable year is dropped: it cannot go into a time bucket and is useless downstream regardless of source.

The second format is the British Library's digitised nineteenth-century collection, read as JSONL pages inside per-decade `.tar.gz` archives. `iter_bl_records` groups pages by volume and takes the year from whichever page in the volume carries a date field; it keeps a volume only if its *first* page is tagged English, not a check of every page in it, so a volume that switches language partway through is not caught. A fixed set of archives, `BL_SKIP_ARCHIVES`, is skipped outright: the pre-1800 decades, since TCP's own transcription already covers that span more cleanly, and `unk.tar.gz`, whose records carry no date at all and would be dropped anyway. Neither TCP nor the British Library supplies clean text on its own. TCP's transcribers resolved most OCR-class problems by hand, which is why Stage 2 barely touches TCP text; the Library's scanned pages carry uncorrected OCR damage that Stage 2 exists to address.

A corpus in neither format can still be added without touching anything downstream of this stage. Already-valid TCP P4 XML needs no code change at all: drop the zip files under `<data\_root>/corpus/tcp/regions/<a-region-name>/<a-source-name>/` and rerun the pipeline, since `discover_sources` finds new regions and sources by scanning the folder tree, not from a hardcoded list. Any other raw format needs one new reader function beside `iter_xml_members`, emitting the same `{region, source, doc_id, year, text}` record the TCP path already writes; `bucket_periods.py` through `metrics.py` and the webapp read only that normalised record and need no change. `docs/overview.pdf`'s Bringing your own corpus section gives the full worked version of this contract.

3 Stage 2: OCR Refinement

`src/ocr_refinement.py` repairs OCR-specific damage in a source's text before it is written to `all_docs.jsonl`. It is meant to run for any source listed in `config.yml`'s `ocr_sources` (today, just `[british, bl_19c]`), but the code that decides whether to call it checks the literal tuple `("british", "bl_19c")` against that list, not general membership in it: adding a new OCR'd TCP-format source to `ocr_sources` would not by itself route it through this stage, contrary to what `config.yml`'s own comment implies. TCP is hand-transcribed and never reaches this stage regardless. The module runs four repair steps in a fixed order.

Layer 1a, line-break hyphen repair, rejoins a page's line-break hyphen where it survived into the plain text as a literal `-` followed by whitespace (`"utrum- que"`). A hyphenated compound never has whitespace between the hyphen and the next letter, so this is a plain regular expression with no wordlist involved, and it cannot mistake a compound

for a spacing artifact. A sampled decade archive showed roughly 28% of pages carrying at least one instance.

Layer 1b, bare-space split repair, handles the harder case where the OCR engine dropped the hyphen entirely and left a bare space ("par ticulars"), with no marker left to detect it by. It merges two adjacent words only if the merged form is a real word in a reference wordlist, and at least one of the two halves is not independently a real word on its own: that second condition is what keeps a phrase such as "well known" untouched, both halves already being real words, at the cost of missing splits where both halves happen to be real words too, such as "in put" for "input". The wordlist is built from that period's own already-extracted TCP text, and only from TCP text: `parse_tcp.py` keeps one running word count per period, updated as TCP documents are read, and hands the British Library loop whichever period's count matches a document's date, or an empty one if none does. TCP ends at 1800, so every period entirely after that date gets an empty wordlist, and `repair_bare_space_splits` returns an empty-wordlist call's text unchanged: Layer 1b does nothing across most of where the British Library's OCR damage actually sits. It was validated by hand in the one period a wordlist existed to test it against, 1790-1810, after temporarily adding a small TCP slice so that bucket would have coverage at all: 142 correct merges confirmed in the first 120 changed documents, following a first validation attempt that turned out to be a false positive, a detection script checking whether two fragments existed anywhere in a document rather than whether they sat adjacent at the claimed merge site. Periods entirely after 1800 remain both unrepaired by Layer 1b and untested.

Layer 1c, long-s misreads, is a named, called function that does nothing. The one OCR'd source in this pipeline postdates the long s's disappearance from English printing, so there is no text anywhere in the corpus to calibrate a character-confusion rule against; a rule guessed without that calibration would risk corrupting every letter *f* in the real text for no benefit. The function stays in place so a future pre-1800 OCR'd source only needs it filled in, not the pipeline rewired around it.

Layer 2, lexicon correction via `analiticcl`, is built, tested, and shipped off. It replaces a garbled token with the closest match in a reference lexicon, within an edit-distance and confidence threshold, and testing found two independent reasons not to turn it on. No confidence-score threshold separates a correct fix, such as NOVEMRER to november, from a wrong one, such as ungodliness to godliness, which reverses the word's meaning: the score ranges for correct and incorrect fixes fully overlap. An affix-aware bypass narrowed this without solving it, fixing five of six documented wrong corrections of that shape while losing none of five documented correct ones, but a larger problem surfaced alongside it. The reference lexicon, MorphAdorner's nineteenth-century fiction corpus, has close to no coverage of proper nouns, technical vocabulary, or legal and civic terms,

so it wrongly corrects such words in every non-fiction genre tested: a natural-history text lost *Sciurus* to *Scaurus*, a sermon lost *Shuhite* to *Shute*, and Coleridge's *Christabel* lost its own title character's name, *Leoline*, to *Leonine*, three times in one document. Across twenty-five documents, even with the affix bypass on, the model fired 14,038 corrections, about one every forty to fifty words, far too dense to be OCR damage. Layer 2 stays in the code, reachable only by constructing a variant model and passing it to `refine()`, since there is no `config.yml` flag for it, because a project with a broader or genre-matched lexicon could reasonably turn it on; nothing in this pipeline does.

`parse_tcp.py`'s `main()` used to do exactly that, unintentionally: it built a Layer 2 variant model and passed it into `refine()` for any British Library period overlapping a TCP source in time, keeping Layer 2 live for those periods against every piece of documentation saying it shipped off. The bug was found the same day Layer 2 was decided against for a second time, while checking the code against the letter of that decision. No production run had executed the path when it was found, so nothing was corrupted, but the next full corpus rebuild would have run it. `main()` now only calls `refine(text, wordlist)`, relying on the function's own `variant_model=None` default instead of constructing one.

4 Stage 3: Bucket into Periods

`src/bucket_periods.py` sorts every extracted document into a twenty-year time window and writes one plain-text file per period, one document per line, plus a matching `<period>_<region>.txt` for every region present. The windows are uniform across the whole 1510 to 1910 span. Fine resolution only around the Sattelzeit would bias the method toward finding a reorganization only where one is already expected, and Heuser's own earlier chapter found a second, unrelated acceleration window around 1905 to 1925 that a Sattelzeit-only resolution could not have detected. The 1510 start, not 1500, is a deliberate ten-year shift: it puts both 1770 and 1830 exactly on a period boundary instead of in the middle of one, so the transitions that open and close the Sattelzeit are each a literal edge between two periods, not an approximation.

This stage's own width, twenty years, has never been stress-tested the way Stage 4's hyperparameters and Stage 5's `top_k` have been (§5, §6). It is one of several settings in this pipeline still resting on a first guess rather than a robustness check, alongside `ocr_refinement.py`'s `min_merged_frequency`, `score_threshold`, and `freq_weight`, and Stage 6's `max_community_size` and `band_width`. Nothing currently points to any of them being a real problem.

5 Stage 4: Train Embeddings

`src/embeddings.py` trains one Word2Vec model per period, and per region-split variant, on that variant's own bucketed text: a two-hundred-dimension vector space in which words that tend to appear near each other end up positioned near each other. Current settings are `window=5`, `min_count=50`, `vector_size=200`, `epochs=10`. A variant with fewer than twenty documents is skipped outright; a handful of idiosyncratic books is not a period-level signal.

These two headline parameters, `min_count` and `window`, went unvalidated until a 2026-08-28 robustness sweep tested four alternative values of `min_count` and three of `window` against eight pilot periods bracketing the Sattelzeit, screening at three of the fifteen mandatory resolutions instead of the full sweep. No alternative produced a cleaner Sattelzeit-localized signal than the defaults, so the defaults were kept; that is the entire content of the decision, and it is easy to misread as a confirmation of the underlying claim. It is not one. The sweep surfaced an unresolved picture: the reorganization signal under every tested configuration rises gradually from early to late periods instead of spiking and returning inside the Sattelzeit window, and shared vocabulary between consecutive periods grows roughly ninefold across the run, tracking the British Library's bulk entry into the corpus in the same late periods that still carry unrepaired OCR noise. Part of the late-period rise in the project's headline metric could be that noise, not semantic reorganization; the sweep neither confirms nor rules that out, and the full fifteen-resolution version of it, which CLAUDE.md's hard constraint requires before any reorganization claim can rest on it, has not been run.

6 Stage 5: Build the Network

`src/network.py` turns one period's embedding space into a graph: every remaining word becomes a node, and an edge connects each word to its `top_k` nearest neighbours by cosine similarity. A fixed similarity threshold was tried first and dropped: at 0.3, the resulting graphs were dense enough to be unusable, roughly ten percent of all possible edges present, tens to hundreds of millions of edges per period at this vocabulary scale, too large for community detection and too large to store. `top_k=15` bounds the graph to roughly fifteen edges per word regardless of vocabulary size, and this value was itself tested on 2026-08-31 against the same eight pilot periods Stage 4's sweep used, at values of 5, 10, 20, and 30, screening at the same three resolutions rather than the full fifteen, the same limitation as Stage 4's sweep and not yet a paper-ready robustness claim on its own. Three of the four alternative values produced the same qualitative picture as the default, a gradual rise rather than a Sattelzeit-localized spike; the fourth, `top_k=5`, sat uniformly higher across every transition, early periods included, which reads as a sparser graph being generally less stable under Leiden, not as specifically

more sensitive to the Sattelzeit.

Before edges are built, two kinds of word are excluded from becoming nodes at all: function words and early-modern grammatical variants (hath, doth, thou, and similar) in a fixed STOPWORDS set, and any single-letter token, which in this corpus is always either a stopword already, a Latin citation abbreviation, an interjection, or an OCR artifact. Neither carries the conceptual content this project measures. This exclusion is incomplete. Roman numerals (xxiiij, xxvij) and Latin citation-apparatus abbreviations (capvt, deute, regu) still pass through into the network from early-modern, hand-transcribed periods, correctly transcribed originals, not OCR damage, and cluster tightly enough by co-occurrence to form their own communities. The labeling stage (§8) already names these accurately, as *Structural / Uncertain* clusters such as *Biblical Book & Chapter Citations*, but the fix belongs here, at the stage that decides what counts as a node, and is not yet built.

Two further capabilities exist in this stage's code and ship off. `use_gpu` offloads the cosine-similarity computation to a GPU via `cupy`; a benchmark against the largest model this pipeline has built, 171,382 words, measured a modest 1.18× speedup, confirming this stage is not today's bottleneck. The capability is banked for a larger future corpus. `pos_filter` restricts which words can become nodes to those tagged noun or adjective, Jamie's proposal for a different problem: without it, several of a period's largest communities are pure grammatical clusters, verb-conjugation groups with no topical content. The one direct before-and-after comparison run so far, on one period, 1750-1770, Combined region, at one resolution, 4.0, confirmed the fix works there, turning grammatical clusters into topical ones, including a vice-words and a virtue-words community that read as near mirror images of each other, but it is not a clean win: removing verb forms also let a new community of pure person-names emerge, a named-entity cluster rather than a grammatical or a topical one. The filter needs a per-period word-to-category lookup table before it can run for a given period; where that table is missing, `network.py` prints a warning naming the period and builds that period unfiltered instead of failing. The table exists for TCP periods through a part-of-speech-tagged release of the same corpus, roughly 95% coverage of this project's vocabulary, lower at 87% for 1790-1810, but the pre-1700 portion of that release has not been downloaded, and the British Library tables are still paused mid-build.

7 Stage 6: Detect Communities

`src/community.py` runs the Leiden algorithm over each period's network to find communities: groups of words more tightly connected to each other than to the rest of the graph. There is no fixed number of communities anywhere in this pipeline; it is always

an output of the algorithm.

Leiden takes a **resolution** parameter trading community size against community count, and every quantitative reorganization claim this project makes has to hold at all fifteen fixed resolution values, 0.1 through 16.0, not just whichever looks cleanest. This is CLAUDE.md's one hard rule: picking a single favourable resolution after seeing the result would let the analyst choose the answer instead of measuring it.

That fifteen-value sweep answers the scientific question but produces partitions useless to show a reader directly, since at low resolution a period's largest community can hold thousands of unrelated words. A second, separate resolution decides what the webapp displays and what the labeling stage reads, picked independently for each variant rather than once globally. On 2026-08-27 an automatic ascending scan replaced the earlier single hand-picked constant, 4.0, that had forced every early, easy period up to whatever the hardest late period needed; the following day, that global scan was itself replaced by the current per-variant band-bisection, since forcing every variant to one shared value carried the same problem the hand-picked constant had, only automated. It works by starting from a seed resolution and doubling or halving until it brackets the lowest resolution at which no community exceeds a fixed 2,500-word cap, then bisecting within that bracket until a passing value lands within 300 words of the cap or the bracket narrows past a safety floor. There is no hardcoded ceiling; the doubling search finds its own upper bound, so a larger future corpus needing resolution 40 instead of 16 does not require anyone to notice and raise a constant by hand.

This mechanism shipped with a bug, caught only once it ran against the full rebuilt corpus rather than the smaller test set it was checked against in 2026-08-28. 1650-1670_american, 840 words total, never exceeds the 2,500-word cap even at the lowest resolution tested, so the bisection routine's search for a resolution that fails the cap never found one, and the code crashed expecting a failing lower bound that was never going to exist. The fix: when no resolution ever fails the cap, the lowest resolution actually tested is used directly, with no bisection needed.

8 Stage 7: Label Communities

`src/extract_community_words.py` and `src/label_communities.py` turn each period's bare Leiden communities into plain-English labels a historian can read, through a CSV-based workflow meant to be run, and corrected, by a person rather than trusted blindly from a single model call.

A community's label is not generated independently every time its period comes up. `generate` walks periods in chronological order and, for each community, first

checks whether the same Hungarian alignment `metrics.py` uses to compute **migration_fraction** (§9) maps it back to a specific predecessor community that already has a label; that check is free, since the alignment itself needs no model call. Whether the inherited label is then actually reused is not free. An earlier version generated each period's label independently from just that period's own top words, and produced a real, publicly visible contradiction: the word *system* was flagged as having *stayed* in the same community between two consecutive periods, correctly, by the same alignment `metrics.py` already trusted, while the community's displayed label changed anyway, from *Books, Manuscripts & Learned Vice* to *Classical Authors & Scholarship*, because the model reading a slightly different word list described it differently. To a reader that looks like the tool contradicting itself. Inheriting the label whenever the alignment says the community is the same makes *the label changed* and *the word moved* the same event by construction, not two independent judgments that can disagree.

Verbatim inheritance on its own then created the opposite failure: with no mechanism to force a fresh read, a label could ride unchanged for a community's entire lifetime while its content drifted completely away from what the label describes. One documented case: a community genuinely holding Dutch and German text at 1510 to 1530 was still labeled *Dutch and German Text* fifteen periods, roughly three hundred years, later, by which point its content had passed through Scots legal prose, classical mythology, and early ecclesiastical history before landing on English church governance, none of it Dutch or German. The current mechanism, and the actual cost this stage pays, is two model reads every time a label would otherwise inherit: both given the same inherited label and the same current word sample, both asked the identical plain yes-or-no question, does this label still fit, independently of each other so that agreement is a real redundancy check and not a comparison between two different questions. Only unanimous agreement skips a fresh read; either reader saying no, or the two disagreeing, forces one immediately.

The word sample a community shows, to both the fresh-labeling prompt and the fit-check prompt, changed on 2026-08-31. It used to be a flat list of the twenty-five words with the highest network degree in that community, its most central members. That sample only ever shows words tightly bound to the community's core, which hides the opposite case: a community with a coherent-looking core can still have a drifting low-degree tail belonging to a different topic, and a reader who never sees that tail has no way to notice. The word list shown is now a stratified sample by degree rank, not a flat top slice: roughly nine words from the top of the ranked list, nine from around its midpoint, and nine from its bottom, drawn from a membership list that can hold up to 2,500 words, not an exhaustive split of it. A community small enough that this would show nothing new, twenty-seven words or fewer, is shown whole instead.

The redo this change required, run across all three regions, caught drift the flat list had been hiding. One community's label, *Latin Liturgical & Hymn Vocabulary*, no longer matched its own core, which had drifted to English virtue-adjectives. A different community kept an OCR-fragment label naming the wrong stripped prefix after its dominant fragment pattern had itself shifted. A third, *Church Fathers & Ecclesiastical History*, had drifted into ordinary English clergy surnames. The aggregate fit-check failure rate did not move the same way in every region: roughly eight percent for Combined against six before the change, sixteen for British against seventeen, and forty for American against fifty, the last on a sample of only 42 checks, too small to read much into on its own. Combined's rise, on 1,310 checks, is not a small-sample artifact, and the project's own working log treats it as evidence that the tiered view catches drift the flat list missed.

Every fresh label names exactly one lane from a fixed list of fifteen, decided 2026-08-03 from 707 labels already produced across earlier runs, instead of inventing a category per community. The fifteenth lane, *Structural / Uncertain*, covers a community grouped by grammar rather than topic, by a shared non-English language, by proper names, or, in the British Library's nineteenth-century periods, by shared OCR damage: a cluster of prefix- or suffix-stripped fragments rather than words at all.

This project's stated reproducibility standard for these labels is what its internal documentation calls Tier 1 or 2: the same input produces a comparable label, not the same label token-for-token, since a model call is not bitwise-deterministic even against an unchanged prompt. It does not claim Tier 3, exact reproducibility.

9 Stage 8: Measure Change

`src/metrics.py` is where the pipeline's central question gets a number. For every pair of consecutive populated periods, at every one of the fifteen mandatory resolutions, it restricts to the vocabulary shared by both periods, since each period's embedding model is trained only on that period's own text, and computes three things: normalized mutual information and adjusted Rand index, two label-permutation-invariant measures of how similar the two periods' community structure is, and **migration_fraction**, the fraction of shared words that end up in a different community even after the best possible one-to-one relabeling between the two periods, found by the Hungarian algorithm on the word-overlap contingency table between every pair of communities. This is the same alignment mechanism Stage 7 uses to decide whether a label inherits, applied here to a different question: not *what this community should be called*, but *how much of the vocabulary structure reorganized*. A pair of periods sharing fewer than twenty words is skipped; a metric computed over that few words would not mean

anything.

10 Region split: Combined, American, and British

Most stages loop over three variants per period, not one: **Combined**, trained on every document regardless of source, and one independently trained pass per region the raw corpus contains. **American** is the Evans collection alone, thin before roughly 1630 and again after 1810, matching Evans's own coverage. **British** pools four archives that never get their own separate variant, EEBO phase 1, EEBO phase 2, ECCO, and the British Library's nineteenth-century supplement, all tagged `british` regardless of which of the four a given document came from.

None of the three variants is a blend of the other two. Combined is not an average of American and British; it is its own independent Word2Vec model, trained on the pooled text directly, the same way American and British are each trained on their own restricted text. The region split answers a provenance question, not a dialect one: does mixing the British Library's OCR'd nineteenth-century text next to TCP's clean, hand-transcribed earlier periods change the reorganization signal in a way that traces back to where the text came from, not to anything the Sattelzeit hypothesis predicts. It is not a claim that American and British English are dialects worth comparing on their own terms; the American variant, in particular, is too thin across most of the corpus's span to support that kind of comparison even had it been intended as one.